

Lead-Lag Origination for Prioritizing Financial KOL Narratives

Yuncong Liu*

yuncongliu@u.nus.edu

National University of Singapore
Singapore

Yao Lu†

luyao@comp.nus.edu.sg

National University of Singapore
Singapore

Abstract

LLM-based financial agents increasingly use external information streams for forecasts, alerts, and trading decisions. Financial social media often reacts before slower channels, but its value is concentrated in the brief window before cascades, resharing, or popularity become observable. Raw-feed ingestion uses that window poorly: LLMs must choose among many weakly separated candidates, raising cost and often lowering selection quality.

Across an approximately 20-million-post financial KOL archive, we identify a stable lead-lag origination structure. Its account roles persist across time and symbols yet are nearly orthogonal to audience size, consistent with distinct roles in financial information production: narrative originators and amplifiers. This structure captures a pre-cascade layer of information intermediation before collective attention becomes visible. We encode posting-time-adjusted account roles and the divergence of temporal-precedence flow in the **Lead-Lag Router (LLR)**, a lightweight point-in-time router for newly originated narratives. LLR consistently prioritizes future high-reach narratives at origination and provides a near-free front end for downstream agents, improving narrative triage and early detection with microsecond-level latency and no token cost.

CCS Concepts

• **Information systems** → **Learning to rank**; *Social networks*; Decision support systems.

Keywords

financial social media, lead-lag networks, narrative routing, learning to rank, financial agents, information diffusion

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1 Introduction

LLM-based trading and forecasting agents increasingly use external information streams [19, 27, 33, 34, 36], shifting part of the bottleneck from reasoning to input selection. Financial social media is attractive because it is early, but it is also noisy, identity-dependent, and useful only briefly; real-time streams remain understudied in LLM trading agents [9]. Market-moving posts can lose value within seconds [26], whereas popularity becomes observable only after

Table 1: Archive scale and structural validation.

Archive scale		Structure validation		
Measure	Value	Test	Estimate	Evidence
Period	2009–2026	Directionality	.716/.618	obs./null; $p = 1/2001$
Posts	19.87M	Cashtag breadth	117/127	BH-FDR 5%
KOLs	3,702	Time stability	$\rho = .447$	$n = 2,104$; $p = 6 \times 10^{-104}$
Co-mention posts	545,002	Shuffle null	95th = .037	–
Events	98,998	Symbol stability	$\rho = .50$	$n = 279$; $p = 5 \times 10^{-19}$
Cashtags	28,146	Follower count	$\rho = -.068$	$n = 641$
Account pairs	24,487	Original posts	16/17	$p < .01$

diffusion. The operational question is therefore which newborn narratives merit reasoning before consensus forms.

Current systems typically read each post, aggregate sentiment, or wait for early popularity. Per-post reading makes raw-feed intake expensive, while popularity-based signals arrive only after reshares and adopters have revealed part of the cascade. Neither directly addresses the earlier selection decision.

Reading the raw feed is costly and can also hurt selection. In our triage study, the same LLM captures less follower-weighted reach from 30 candidates than from a router-produced shortlist of ten because the larger pool contains more near-duplicates and weak signals. It also raises token cost and gives a noisy selector more opportunities to choose the wrong item. We therefore place a point-in-time router before expensive reasoning; Figure 1 shows how it filters KOL narratives while leaving the downstream agent unchanged.

At origination, such a router can use only the originator, event context, and prior account history. We ask whether financial Key Opinion Leader (KOL) streams contain a persistent signal of who introduces narratives before collective attention forms. The archive reveals stable posting-order roles across time and symbols that are nearly orthogonal to follower count, separating narrative origination from attention amplification. Some accounts repeatedly enter asset discussions before others: a small account can lead, while a large central account can still follow. In Figure 2, arrows encode first-post order in prior co-mention events rather than follow links or retweets; the inferred role also adjusts for habitual posting hour.

We validate this *lead-lag origination structure* at two scales (Table 1). A 2009–2026 archive contains 19.87M posts from 3,702 KOLs. Structural estimation uses 545,002 co-mention posts forming 98,998 cashtag-day events and 24,487 recurrent account pairs. Pairwise directionality is .716, versus .618 after within-event time shuffling, and is significant for 117/127 measurable cashtags. On a dense 17-symbol panel,¹ account roles correlate .447 across adjacent windows and .50 across disjoint symbol sets, far above the shuffle null, but only $-.068$ with follower count. The hierarchy also remains significant for original posts in 16/17 symbols, indicating that resharing alone does not create the pattern.

*Asian Institute of Digital Finance.

†School of Computing. Corresponding author.

¹The panel comprises AAPL, MSFT, NVDA, TSLA, AMZN, META, GOOGL, AMD, MSTR, COIN, HOOD, PLTR, SPY, QQQ, BTC, ETH, and SOL.

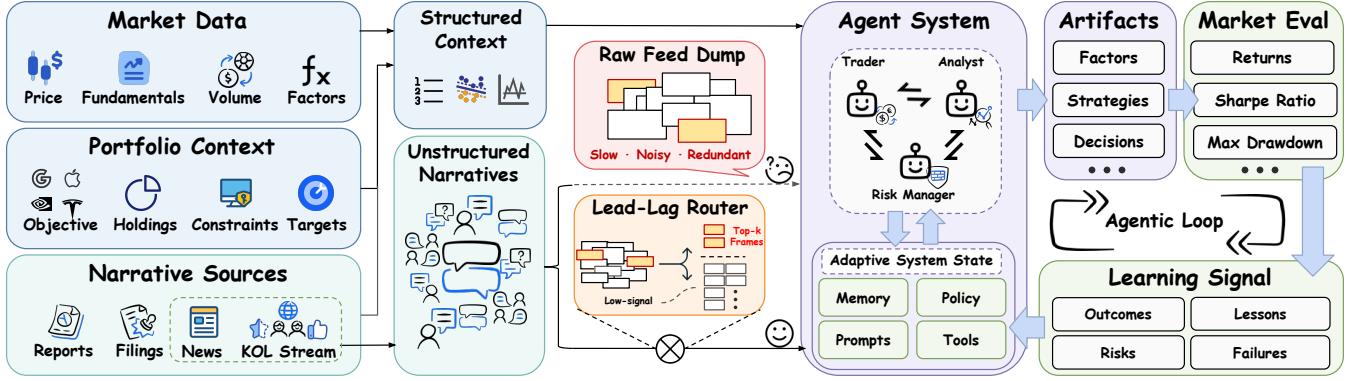


Figure 1: LLR as a narrative-intake layer for financial agents. Structured market and portfolio context enters the agent directly; LLR routes newly originated KOL narratives before downstream reasoning, leaving the rest of the agentic loop unchanged.

These results identify a pre-cascade layer of financial information intermediation in which persistent early entry differs from audience reach, without implying private information or causal price discovery. Audience size proxies amplification capacity, whereas the lead-lag role captures repeated early entry; the accounts introducing a view need not be those that later make it visible. The **Lead-Lag Router** (LLR) encodes this structure in a point-in-time ranker that uses no post text after frame formation. It ranks newborn narratives by future follower-weighted reach and serves unchanged downstream systems through narrative ranking, LLM triage, and early attention-breakout detection.

Contributions.

- We identify and archive-scale validate a persistent financial narrative-origination structure distinct from audience size.
- We show that degree-normalized temporal precedence, rather than centrality or popularity, carries the useful graph signal and that identity shuffling removes the gain.
- We introduce LLR, a microsecond, text-free router that outperforms the evaluated encoders and LLMs and improves triage and early-event detection.

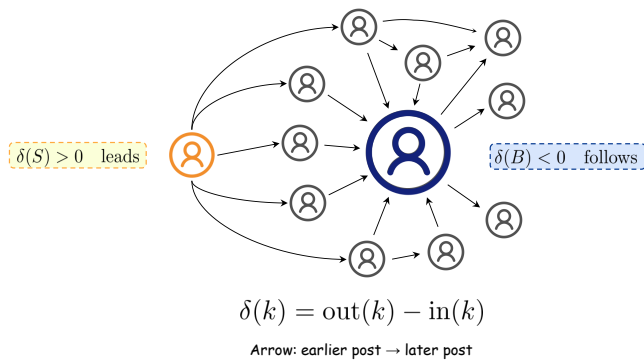


Figure 2: Lead-lag flow as net temporal precedence.

2 Related Work

Financial Agents and Intake Routing. Financial agent systems increasingly integrate market data and external information into forecasts and trading decisions [19, 27, 33, 34, 36]. Related systems optimize execution across dependent agent calls or route queries to reward models [29, 32]; cost-aware LLM cascades and retrieval pipelines likewise narrow inputs before expensive reasoning [8, 10]. LLR instead routes financial narratives themselves, allocating scarce analytical attention before diffusion using historical origination structure rather than query–document relevance or model confidence.

Signals in Financial Social Media. Financial social media both produces market interpretations and allocates attention. Text, discussion, and sentiment carry signals about market outcomes [2, 17, 24, 28], while influencer identity, analyst recommendations, and expert track records also matter [4, 12, 14, 38]. These studies ask what is said, who commands attention, or whose advice succeeds; recent work also studies policy completion from financial KOL discourse [20]. LLR models the pre-diffusion division between accounts that introduce asset narratives and those that amplify them as financial information intermediation.

Cascades and Popularity Prediction. Cascade models predict popularity from observed or modeled diffusion. DeepCas learns from early cascade structure [18], whereas CasMS spans a zero-observation generation stage and later diffusion stages [37]. At generation, CasMS still uses message content and static social-graph position; later stages add diffusion evidence. LLR instead scores a newly originated frame without text or observed diffusion.

Influence, Precedence, and Graph Operators. Existing influence measures capture several account properties. Follower count is a weak proxy [7]; PageRank and HITS score recursive connectivity [16, 22]; Romero et al. model influence and passivity [25]; stable-influencer work tracks persistent success [35]; and node2vec learns latent position from random-walk neighborhoods [11]. The closest structural ancestor is iRank [1], which builds a temporal-precedence graph over shared referents but scores it with PageRank-style centrality; Menezes et al. likewise classify precursors and laggards

without a forward prediction target [21]. Building on this line, LLR replaces descriptive centrality with a degree-normalized divergence operator and uses it as a forward, point-in-time financial-routing signal. We validate the resulting origination structure at archive scale in financial KOL streams (Table 1). On the same graph, the net-lead rate outperforms centrality scores (§5.4). Centrality measures access to attention; divergence measures repeated precedence in introducing asset narratives.

Directional Ranking and Hodge Structure. HodgeRank identifies divergence as the source term of an antisymmetric comparison flow [13], and Hodge methods have been applied to financial return networks to recover causal hierarchy [30]. LLR degree-normalizes this source term, yielding a one-step local approximation to the global potential for pre-cascade financial information intake rather than a standalone descriptive ranking.

3 Problem Formulation

Given this role separation, a financial agent faces an operational decision: should it spend scarce reasoning capacity on a new asset narrative? Before any cascade forms, we cast this as point-in-time narrative ranking.

3.1 Point-in-time narrative ranking

Let $\mathcal{K}$ be the set of KOL accounts. A post is (k, τ, c, x) , with author k , timestamp τ , symbol c , and text x . An event is a symbol-day pair $e = (c, d)$ with at least $\kappa=8$ participating KOLs. Within an event, streaming novelty detection groups posts into frames (§4.1). A frame f is the routing unit, with origination time $\tau_0(f)$ and originator $k_0(f)$ given by its earliest post.

The label is same-day future follower-weighted reach:

$$R(f) = \sum_{k \in \text{adopt}(f)} \text{fol}(k), \quad (1)$$

$$\text{adopt}(f) = \{k \neq k_0(f) : \tau_k(f) > \tau_0(f)\},$$

where $\tau_k(f) = \min\{\tau : (k, \tau) \in f\}$. We rank by $\log(1+R(f))$. This label measures subsequent reach within the financial attention network, not price impact or trading profitability; it is realized only after $\tau_0(f)$ and is unavailable at routing time.

The point-in-time constraint is

$$\phi(f) \in \sigma(k_0(f), C_0(f), H_{<B}(k_0(f))),$$

where $C_0(f)$ is context at $\tau_0(f)$ and $H_{<B}(k)$ is account history before the evaluation block. The ranker may not use later adopters, reshares, partial popularity, or future same-day posts. Each event contributes its first $K=10$ frames by origination order, scored by

$$s_\psi : \phi(f) \mapsto \mathbb{R}, \quad \phi(f) = (\phi_{\text{ctx}}(f), \phi_{\text{orig}}(f)). \quad (2)$$

Evaluation uses NDCG@3 and Hit@1 as primary top-of-list metrics. Text embeddings form frames, but neither ϕ_{orig} nor the final scorer reads post text (§4.2, §4.3).

3.2 Why routing precedes reasoning

A financial agent has bounded reasoning capacity, so feeding the whole candidate pool to an LLM is not a neutral baseline. A larger pool gives a noisy selector more opportunities to choose a non-best

item. Model any selector as

$$\hat{s}_i = v_i + \varepsilon_i, \quad \hat{b} = \arg \max_i \hat{s}_i,$$

where v_i is true value, ε_i is independent zero-mean σ -sub-Gaussian noise, and $b^* = \arg \max_i v_i$. For any candidate set S containing b^* , let $\text{cap}(S) = \Pr[\arg \max_{i \in S} \hat{s}_i = b^*]$.

Proposition 1 (Pre-filtering and selector capacity). *For a fixed target, $\text{cap}(S) \geq \text{cap}(P)$ whenever $b^* \in S \subseteq P$. If a pre-filter returns $\mathcal{F} \subseteq P$ before the selector noise is drawn (equivalently, independently of that noise conditional on candidate values), with recall $\rho = \Pr[b^* \in \mathcal{F}]$, then*

$$\text{cap}(\text{filter} \rightarrow \text{selector}) = \rho \Pr[\hat{b} = b^* \mid b^* \in \mathcal{F}] \geq \rho \text{cap}(P). \quad (3)$$

Proof sketch. Passing from S to P intersects the success event with additional pairwise comparisons, so $\text{cap}(S) \geq \text{cap}(P)$. Conditioning on $b^* \in \mathcal{F}$ and using the stated noise independence then gives Eq. (3).

The sub-Gaussian maximal inequality gives the noise scale

$$\mathbb{E} \left[\max_{i \leq n} \varepsilon_i \right] \leq \sigma \sqrt{2 \ln n}$$

(Thm. 2.5 of [5]). For Gaussian errors, where this logarithmic growth is tight, balancing it against a best-item margin Δ suggests the characteristic capacity

$$n^* \approx \exp(\Delta^2 / 2\sigma^2). \quad (4)$$

For $\rho > 0$, filtering improves end-to-end selection exactly when $\Pr[\hat{b} = b^* \mid b^* \in \mathcal{F}] > \text{cap}(P)/\rho$.

The archive-scale structure summarized in Table 1 supplies a pre-cascade originator signal for this filter; §4.2 converts it into routing features.

4 Methodology

LLR turns the role separation established in Table 1 into the point-in-time gate of §3. Each newborn frame combines origin-time context with Ω , its originator's persistent early-entry role, and g , its relational precedence among co-attending accounts. A zero-text listwise model ranks the frames. Figure 3 and Algorithm 1 show the pipeline; every feature for block B uses only posts before its cutoff t_B .

4.1 Frame construction

For each symbol-day event (§3), we keep each KOL's earliest post. Frames are built by a streaming novelty pass [23]: MiniLM embeddings [31] are merged into the nearest existing frame if cosine similarity exceeds $\theta=0.50$, otherwise they open a new frame. We keep the first $K=10$ frames per event, labeled by same-day reach (Eq. (1)). MiniLM at $\theta=0.50$ is the anchor labeler; the main experiment also uses Qwen3-Emb-4B and BGE-base thresholds chosen to match frame granularity.

4.2 Lead-lag graph features

The account-level role Ω_k summarizes persistent early entry. Within an event e , we order participating accounts by first-post time and assign account k the net lead $\ell_e(k) = m_e + 1 - 2 \text{rank}_e(k)$, where m_e

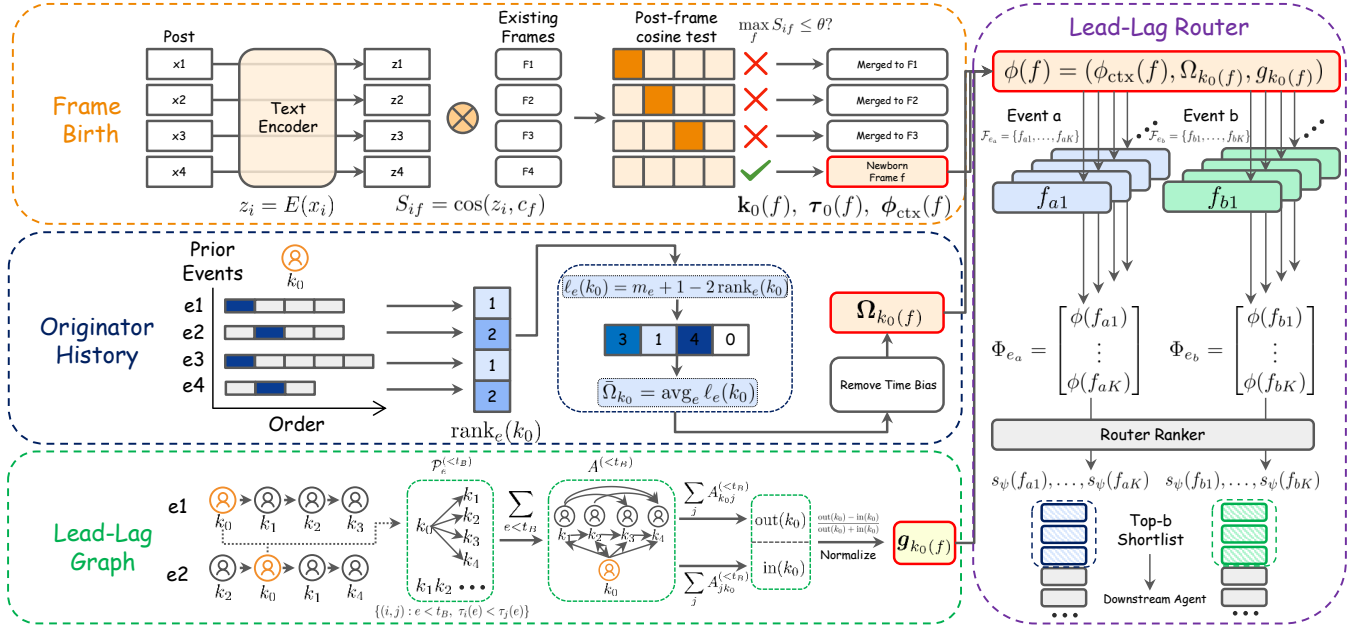


Figure 3: Lead-Lag Router mechanism: a streaming novelty test forms newborn frames, whose origin-time context and historical originator signals (Ω , g) feed a listwise ranker that emits an event-wise shortlist for downstream reasoning.

Algorithm 1: Lead-Lag Router

Input: Stream $\mathcal{P}$; temporal blocks $\{B\}$; thresholds κ, θ ; cap K

Output: Trained zero-text ranker s_ψ

```

1 foreach block  $B$  with cutoff  $t_B$  do
2    $\mathcal{H}_B \leftarrow \{p \in \mathcal{P} : \tau(p) < t_B\}$ 
3    $A_B \leftarrow [A_{ij}^{(<t_B)}]$  by Eq. (5)
4    $\Omega_k^{(B)}, g_k^{(B)} \leftarrow \text{LeadLagFeatures}(\mathcal{H}_B, A_B, k)$ , with
     missing accounts set to 0
5   foreach  $e \in B$  such that  $|\mathcal{K}_e| \geq \kappa$  do
6      $\mathcal{F}_e \leftarrow \text{Novelty}_\theta(e)$ ,  $C_e \leftarrow \text{first}_K(\mathcal{F}_e)$ 
7     foreach  $f \in C_e$  do
8        $\phi_f \leftarrow (\phi_{\text{ctx}}(f), \Omega_{k_0(f)}^{(B)}, g_{k_0(f)}^{(B)})$ ,
9        $y_f \leftarrow \log(1 + R(f))$ 
10    end
11 end
12  $s_\psi \leftarrow \arg \min_\psi \sum_e \mathcal{L}_{\text{XENDCG}}(\{s_\psi(\phi_f), y_f : f \in C_e\})$ 
13 return  $s_\psi$ 

```

is the number of participants. We average $\ell_e(k)$ over the account's prior events $\mathcal{E}_k^{(<t)}$ (at least four):

$$\bar{\Omega}_k = |\mathcal{E}_k^{(<t)}|^{-1} \sum_{e \in \mathcal{E}_k^{(<t)}} \ell_e(k).$$

Because $\bar{\Omega}_k$ correlates with median UTC posting hour ($\rho = -.77$), Ω_k residualizes it against a quadratic in that hour. It therefore

measures historical temporal precedence, not textual novelty, forecasting skill, or private information.

For cutoff t , define the historical directed adjacency

$$A_{ij}^{(<t)} = \sum_{e: e < t} \mathbf{1}\{i, j \in e\} \mathbf{1}\{\tau_i(e) < \tau_j(e)\}, \quad (5)$$

where $\tau_i(e)$ is account i 's first post time in event e . Let $\text{out}(k) = \sum_j A_{kj}^{(<t)}$ and $\text{in}(k) = \sum_j A_{jk}^{(<t)}$. Their difference $\delta(k) = \text{out}(k) - \text{in}(k)$ is positive when k historically leads its co-attending accounts and negative when it follows.

The two signals read historical origination at different scales (the middle and bottom lanes of Figure 3). Ω_k summarizes an account's persistent early-entry role across events, whereas g_k measures its net precedence relative to co-attending accounts:

$$g_k = \begin{cases} \frac{\delta(k)}{\text{out}(k) + \text{in}(k)}, & \text{out}(k) + \text{in}(k) > 0, \\ 0, & \text{out}(k) + \text{in}(k) = 0. \end{cases} \quad (6)$$

Accounts with no prior lead-lag edges also receive $\Omega_k = 0$. The normalization prevents posting volume from masquerading as origination: g is nearly volume-orthogonal ($\rho = 0.010$ with total activity), whereas raw divergence is activity-confounded ($\rho = -0.21$).

Proposition 2 (Net-lead rate as a one-step HodgeRank potential). *Let $L = D - W$ be the weighted comparison Laplacian with $w_{ij} = A_{ij}^{(<t)} + A_{ji}^{(<t)}$ and $D_{kk} = \text{out}(k) + \text{in}(k)$. With normalized pairwise preferences, the HodgeRank normal equations reduce to $Ls = \delta$, whose solution is $s^* = L^+ \delta$. Starting Jacobi iteration from $s^{(0)} = 0$ gives, for non-isolated nodes,*

$$g = D^{-1} \delta = s^{(1)}, \quad (7)$$

with $g_k = 0$ assigned to isolated nodes.

Here the weighted divergence at node k is exactly $\text{out}(k) - \text{in}(k) = \delta(k)$, so g_k is a diagonally preconditioned one-step approximation to the global precedence potential. Empirically g and $L^+\delta$ correlate 0.98, and the full solve does not improve ranking. PageRank and HITS remain graph baselines because they measure recursive connectivity rather than divergence.

4.3 Router training and deployment

Each candidate frame is represented by

$$\phi(f) = (\phi_{\text{ctx}}(f), \Omega_{k_0}(f), g_{k_0}(f)),$$

where ϕ_{ctx} contains origin-time activity, audience, and timing controls; Ω_k and g_k add the originator’s persistent and relational roles. No post text is used after frame construction.

We train s_ψ as a LightGBM ranker [15] with the listwise XENDCG objective [6]. Each event is a query group; relevance grades are 32-quantile bins of realized log-reach. Hyperparameters are selected by an inner temporal split within the training window. At scoring time, LLR evaluates a shallow tree ensemble over precomputed scalars, at microsecond latency and zero token cost.

The same score supports two downstream uses: a budgeted gate that forwards only a shortlist to an expensive reasoner, and a standalone detector of large-reach events. Both are evaluated as application studies.

5 Experiments

We evaluate LLR on the point-in-time ranking task in §3 and use the same score for LLM triage and early event detection. All results come from archived files released with our code; internal variants appear only in the ablation study.

5.1 Experimental Setup

The downstream benchmark is a dense 17-symbol subset of the archive, containing 4.6×10^5 posts from 641 financial KOL accounts over large-cap equities, index ETFs, and major cryptocurrencies. Following §3, we retain events with at least $\kappa=8$ KOLs and rank their first $K=10$ frames by same-day log follower-weighted reach.

We compare three calibrated frame constructors: MiniLM@0.50, Qwen3-Emb-4B@0.55, and BGE-base@0.78. Their thresholds calibrate encoder-specific cosine scales to about five frames per event; they are not ranking hyperparameters. MiniLM is the lightweight anchor for subsequent studies; early streaming first-story detection reported similar threshold robustness [23].

We report NDCG@3 and Hit@1 for top-of-list accuracy, Mass@3 for captured follower-weighted reach, and JS divergence between predicted and realized attention distributions; smaller JS indicates closer distributional agreement. Applications and ablations use five rolling one-year windows from 2021-06 to 2026-06, with symbol-balanced bootstrap intervals where noted.

Protocol. All features and hyperparameters use pre-test data only. For the main table, we fit before 2025-01, validate on 2025-01–2025-06, refit on 2024-06–2025-06, and test on 2025-06–2026-06; all ranges are left-closed and right-open. The test set has 2,304 symbol-day events and 12,489 frames. Frame thresholds are calibrated only on training data, and block-level account and graph features use posts before the cutoff (§4.2). All trainable baselines

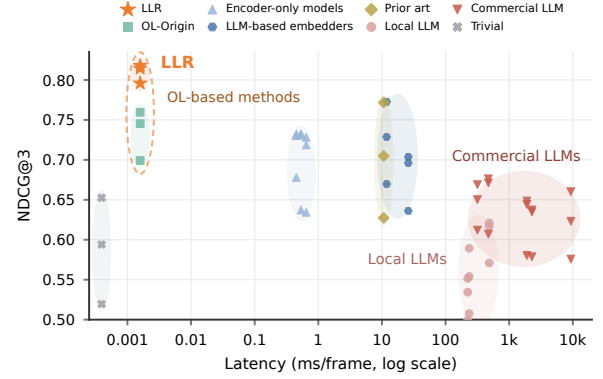


Figure 4: Quality–latency frontier over the three frame constructors.

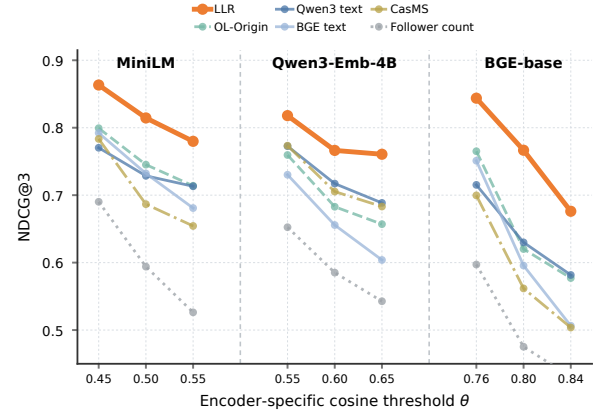


Figure 5: Sensitivity to encoder-specific frame-construction thresholds.

use the same temporal splits. OL-Origin is a zero-text pointwise ridge regressor of $\log(1 + R(f))$ on LLR’s feature vector $\phi(f) = (\phi_{\text{ctx}}(f), \Omega_{k_0}(f), g_{k_0}(f))$; other trainable baselines share LLR’s listwise ranker, while LLMs score candidates directly. Detailed configurations are released with our code.

5.2 Main Results

Table 2 compares LLR with direct LLM scoring, text encoders, and prior social-media baselines under all three frame constructors. Internal LLR variants are deferred to §5.4.

LLR leads all twelve cells, with every gain-row difference significant under a paired symbol-balanced bootstrap ($B=4000$, 90% level). Its lead over text-reading and cascade baselines indicates that historical origination roles are informative before financial attention becomes visible. Figure 4 adds online latency: LLR scores precomputed features in .0016 ms per frame, roughly four orders faster than large encoders and five to six orders faster than the tested LLMs, while retaining the highest NDCG@3. Latency uses batch

Table 2: Point-in-time ranking on the 2025-06–2026-06 test window. N=NDCG@3, H=Hit@1, M=Mass@3; lower JS is better. Bold marks LLR; underlining marks the best baseline without origin structure. Gains use rounded values and report relative reduction for JS.

Family	Method	MiniLM@0.50				Qwen3@0.55				BGE@0.78			
		N↑	H↑	M↑	JS↓	N↑	H↑	M↑	JS↓	N↑	H↑	M↑	JS↓
Trivial	random	.545	.230	.679	.319	.560	.261	.698	.313	.481	.207	.610	.355
	follower count	.594	.310	.713	.330	.652	.345	.782	.303	.520	.268	.653	.377
Local LLM	Llama3.1-8B	.534	.238	.665	.490	.551	.253	.690	.493	.500	.239	.629	.507
	Qwen2.5-7B	.554	.257	.684	.483	.589	.285	.732	.470	.508	.237	.640	.502
	Gemma3-12B	.621	.322	.737	.434	.617	.318	.745	.437	.571	.288	.703	.459
Commercial LLM	Claude Sonnet 5	.667	.400	.779	.394	.664	.372	.786	.399	.581	.318	.711	.447
	DeepSeek V4 Flash	.623	.366	.727	.417	.660	.386	.775	.397	.576	.302	.708	.460
	Xiaomi MiMo-v2.5	.637	.367	.742	.407	.635	.351	.756	.419	.579	.325	.691	.435
	Minimax-M3	.648	.381	.757	.408	.644	.364	.765	.409	.580	.322	.697	.440
	Gemini 3.1 Flash Lite	.669	.390	.769	.387	.650	.369	.772	.410	.612	.344	.732	.424
	GPT-5.4	.671	.409	.775	.384	.676	.399	.787	.383	.607	.361	.717	.414
Prior art	node2vec [11]	.492	.185	.625	.331	.555	.233	.704	.310	.513	.222	.656	.352
	IP algorithm [25]	.557	.239	.676	.317	.548	.249	.693	.310	.535	.293	.642	.352
	Stable spreader [35]	.680	.387	.788	.303	.685	.400	.807	.295	.599	.316	.722	.335
	Expert track record [38]	.682	.393	.779	.305	.690	.406	.807	.299	.611	.325	.734	.340
	CasMS [37]	.705	.444	.800	<u>.272</u>	.772	.539	<u>.855</u>	<u>.245</u>	.627	.352	.745	.324
Text encoder	E5-Mistral-7B	.696	.424	.788	.313	.704	.451	.803	.313	.636	.379	.739	.354
	FinBERT [3]	.731	<u>.450</u>	.820	.276	.732	.473	.832	.287	.637	.376	.742	.336
	Qwen3-Embedding-4B	.729	<u>.444</u>	.819	.285	<u>.773</u>	<u>.546</u>	.853	.251	.670	.408	<u>.778</u>	.316
	BERT-base	.728	.438	<u>.825</u>	.277	.719	.461	.823	.297	.634	.353	.752	.346
	BGE-base	<u>.732</u>	.449	.819	.274	.730	.483	.825	.278	<u>.678</u>	<u>.412</u>	.770	<u>.309</u>
Zero-text	OL-Origin ridge	.745	.436	.836	.274	.760	.489	.860	.268	.699	.417	.819	.309
Ours	<i>Gain vs underlined</i>	<i>+11%</i>	<i>+26%</i>	<i>+7%</i>	<i>+21%</i>	<i>+6%</i>	<i>+10%</i>	<i>+4%</i>	<i>+17%</i>	<i>+17%</i>	<i>+31%</i>	<i>+14%</i>	<i>+20%</i>
	LLR	.815	.565	.885	.214	.818	.602	.886	.203	.796	.541	.885	.246

per-query means for text encoders and CasMS ($n=128$), completed serial runs for local LLMs, batch cache records for commercial models, and the measured origin-feature path for LLR.

Stability checks. LLR remains first across all nine encoder-specific threshold settings; increasing θ creates finer, harder candidate sets and lowers all methods (Figure 5). The ordering also holds over $\kappa \in \{6, 8, 10\}$ and $K \in \{5, 10, 15\}$ (20/20 paired CIs exclude zero). Across three-month to full-history lookbacks, pooled NDCG@3 and Hit@1 vary by less than .01 (Figure 6), so recent activity suffices. All 18 asset-class gains are positive; 90% CIs exclude zero in 6/6 equity, 5/6 ETF, and 2/6 crypto comparisons, consistent with more stable equity origination roles. Only 0.04% of originators fall below the history gate and default to $\Omega=g=0$.

5.3 Application Studies

We next test two financial-monitoring decisions: allocating limited reasoning across simultaneous narratives and flagging those likely to dominate subsequent attention.

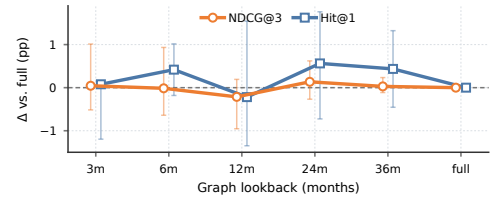


Figure 6: Lookback stability of the normalized net-lead rate g_k , relative to full-history performance.

LLM triage: routing before reasoning. For this application, we use each eligible symbol-day’s first 30 frames, lifting the main study’s $K=10$ cap. Across 200 days in five rolling windows, six LLMs select three narratives from the raw pool or an LLR top-10 shortlist (Figure 1); the DeepSeek sweep also tests arrival-order and random shortlists. Capture is selected follower-weighted reach divided by the event’s oracle top-three reach.

LLR shortlisting improves all six models by $+0.072$ to $+0.134$ (Table 3; Figure 7); the DeepSeek gain is significant at the day level (90% CI $[+.101, +.168]$). The budget sweep in Figure 8 matches Proposition 1: capture falls as the shortlist grows toward the full pool ($.535 \rightarrow .464 \rightarrow .401$); arrival-order (.372) and random (.350) shortlists do not recover it. At $b=10$, LLR also beats the best semantic (.502) and account/history (.428) shortlisters; at $b=20$, it converges with the strongest semantic alternatives. Without an LLM, LLR’s own top three capture .597, as the downstream LLM is not optimized to preserve LLR’s reach-based ordering.

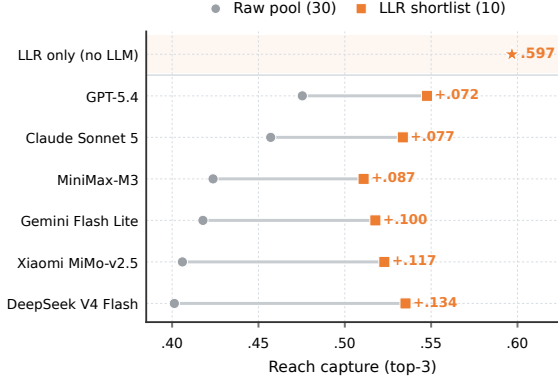


Figure 7: Paired triage shifts across commercial LLMs (top-3 reach capture).

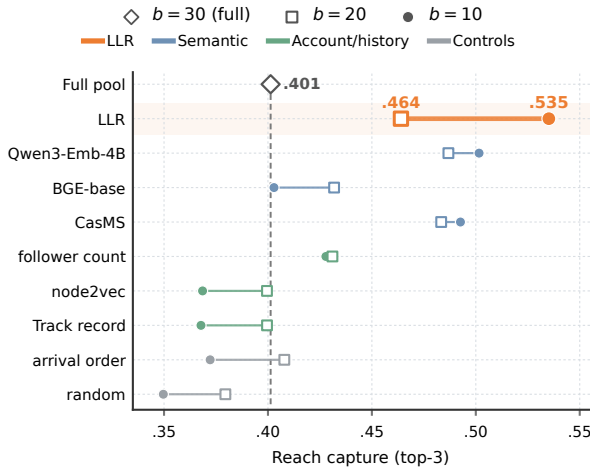


Figure 8: Dilution dose-response under DeepSeek V4 Flash.

Early attention-breakout detection. At the first post, the standalone detector predicts whether a newborn narrative will enter the symbol’s top reach decile, before follow-on posts, reshares, or adopters. Table 4 reports PR-AUC and ROC-AUC over 80 eligible symbol-window cells from the same five rolling windows.

LLR reaches .333 PR-AUC and .790 ROC-AUC. Its PR-AUC lead is significant over Qwen3-Embedding-4B and CasMS (90% CIs: $+.065$

Table 3: LLM triage over five rolling windows (200 decision days). $\dagger$: day-level paired 90% bootstrap CI $[+.101, +.168]$.

Downstream LLM	Raw burst (30)	+LLR shortlist (10)	Δ
GPT-5.4	.475	.548	+.072
Claude Sonnet 5	.457	.534	+.077
Minimax-M3	.424	.511	+.087
Gemini 3.1 Flash Lite	.418	.518	+.100
Xiaomi MiMo-v2.5	.406	.523	+.117
DeepSeek V4 Flash	.401	.535	+.134 $\dagger$
LLR-only top-3 (no LLM)	—	.597	—

$[+.040, +.093]$ and $+.081$ $[+.056, +.107]$); R-precision preserves the ordering (.344 vs. .282). Text-only LLM detectors peak at .207 PR-AUC: content at birth reveals less than LLR’s point-in-time structural signal.

Table 4: Early attention-breakout detection over five rolling windows (80 eligible symbol-window cells).

Method	PR-AUC $\uparrow$	ROC-AUC $\uparrow$
Expert track record [38]	.103	.500
random	.118	.495
node2vec [11]	.123	.482
follower count	.153	.589
BGE-base	.168	.576
CasMS [37]	.254	.711
Qwen3-Embedding-4B	.269	.714
Local LLM		
Local LLM detectors (range)	.122–.154	.513–.596
Commercial LLM		
Xiaomi MiMo-v2.5	.153	.593
DeepSeek V4 Flash	.168	.605
Minimax-M3	.182	.618
Claude Sonnet 5	.189	.644
Gemini 3.1 Flash Lite	.201	.647
GPT-5.4	.207	.665
LLR	.333	.790

5.4 Ablation and Mechanism

Table 5 tests the structure terms, the account–topology assignment, and divergence against centrality.

Table 5: Ablations over five rolling windows (MiniLM@0.50). N =NDCG@3, H =Hit@1; deltas use each block’s baseline. $\dagger$: symbol-balanced 90% CI excludes zero.

Variant	$N\uparrow$	$H\uparrow$	ΔN	ΔH
<i>Structure add-ons, vs. PIT controls</i>				
PIT controls	.781	.514	—	—
+ account role Ω_k	.783	.517	+.002	+.003
+ graph rate g_k	.785	.530	+.003	+.016 $\dagger$
LLR: $\Omega_k + g_k$	.787	.531	+.005	+.017
<i>Identity validity, vs. real LLR</i>				
Shuffled topology	.769	.502	-.017 $\dagger$	-.029 $\dagger$
<i>Graph operator, vs. PIT+Ω_k</i>				
PageRank	.784	.519	+.003	+.001
HITS hub	.783	.519	+.002	+.002
Influence-passivity	.781	.516	-.001	-.002
net-lead rate g_k	.786	.534	+.005	+.016 $\dagger$

Both structure terms improve point-in-time controls, with g_k supplying most of the Hit@1 gain (+.016, the only individually significant increment). Shuffling the account–topology assignment trails real LLR by .017 NDCG and .029 Hit and falls below its matched no-graph control (−.007 NDCG, 90% CI [−.012, −.002]), tying the gain to identity-specific structure. On the same graph, PageRank, HITS, and influence–passivity add nothing, whereas g_k does. Its row is an independent run, explaining the slight difference from the LLR row above. Centrality captures prominence in financial attention, whereas g_k captures repeated precedence in introducing asset narratives.

6 Discussion and Future Work

Case in point. The origination role is distinct from audience size: in 28.8% of the test events where LLR’s top pick is the realized top-reach frame, the winning originator has fewer followers than the event’s median candidate. One ETH event in July 2025 isolates this—an account in the 23rd follower percentile originated the frame and ranked first by Ω , and LLR placed it first at birth while the follower-count baseline and both text encoders placed it fourth; it then drew 18 adopters and 96% of the event’s reach. Content and popularity missed it at the moment that mattered; originator-aware structural features identified it.

Boundaries. The benchmark uses a curated KOL pool, which excludes cold starts by construction; three months is the shortest history validated here, and true cold starts remain open. The lead is strongest in equities and remains positive but less consistently significant in ETFs and crypto. LLR predicts diffusion, not returns: it decides which narratives deserve reasoning and does not replace the trading judgment downstream. The role is a property of current posting behavior; if origination-based routing became widespread, the window it exploits could compress.

Future work. Plugging the router into a full agent loop would test whether earlier, cleaner narrative intake improves portfolio-level outcomes rather than selection capture alone. Rebuilding the lead-lag graph on other platforms and languages would test whether stable origination roles are a general property of financial attention networks. A central economic question is whether persistent originators produce information or broker attention.

7 Conclusion

Financial agents must decide, at the instant a narrative appears, whether it deserves reasoning—before any cascade, reshare, or popularity exists. We showed that financial KOL streams contain a stable lead-lag origination structure: some accounts repeatedly introduce asset narratives before others, nearly independently of audience size, consistent with distinct origination and amplification roles. Encoding this structure as degree-normalized divergence yields the Lead-Lag Router, a zero-text ranker that outperforms all evaluated baselines at microsecond cost. Ablations tie the gain to real account–topology alignment and temporal precedence rather than centrality. Across narrative ranking, LLM triage, and early attention-breakout detection, LLR provides a low-cost intake layer for financial agents.

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